DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

INTERNSHIP REPORT

ON

"AI Automation"

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Chandrahas B - 1DS20AI021

Conducted at

ElementalsAI

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangelore-78

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Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

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CERTIFICATE

This is to certify that the Internship titled **“AI automation”** carried out by **Mr CHANDRAHAS B [1DS20AI021]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

Signature of Reporting Head	Signature of Internship Coordinator	Signature of HoD
Aneesh Nagaraj	**Prof. Kavya D N**	**Dr. Vindhya P Malagi**
Team Lead	Dept. of AI & ML	Dept. of AI & ML
Sandlogic Technologies	DSCE	DSCE

External Viva:

Name of the Examiner	Signature with Date
1)	________________
2)	_________________

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning

D E C L A R A T I O N

I, **Chandrahas B,** final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **ElementalsAI**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023-2024.

Date: USN: 1DS20AI021

Place: Bangalore NAME: Chandrahas B

OFFER LETTER

Internship Offer with ElementalsAI

Suraj Donthi
Founder, ElementalsAI
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Basavanagudi
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Ph. No.: +91-8553406309
Email: suraj.donthi@elementals.ai

Date: 07th March 2024

Chandrahas B
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JP Nagar 1st Phase, Sarakki
Bangalore - 560078
Ph. No.: +91-8105690727
Email: chandrahas.baragur@gmail.com

Dear Chandrahas,

I hope this letter finds you well. I am writing to extend an updated offer of employment to you for the position of AI Research Intern. At ElementalsAI, we believe that our team is our biggest strength, and we
take pride in hiring ONLY the best and the brightest. We are confident that you would play a significant role in the overall success of the venture and wish you the most enjoyable, learning packed, and truly meaningful internship experience with ElementalsAI.

Your appointment will be governed by the scope of work elaborated in Annexure A and terms and conditions presented in Annexure B.

We look forward to you joining us. Please do not hesitate to call us for any information you may need.

Also, please sign the duplicate of this offer as your acceptance and forward the same to us.

Congratulations!!

Suraj Donthi
Founder
ElementalsAI

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr Suraj Donthi**, CEO of ElementalsAI, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N,** Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

CHANDRAHAS B [1DS20AI021]

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CHAPTER 1
INTRODUCTION

Introduction to AI Automation

AI automation combines artificial intelligence technologies with automated processes to streamline tasks across industries. By leveraging algorithms and machine learning, AI automation systems replicate human cognitive functions to analyze data, make decisions, and execute actions efficiently. These systems handle tasks ranging from data analysis to customer service interactions, driving operational efficiency and cost reduction. By reallocating human resources to strategic endeavors, AI automation fosters innovation and growth while reshaping industries and defining the future of work.

1. **Natural Language Interaction.**

One of the key features of AI automation is their ability to understand and respond to natural language queries. Users can communicate with these assistants in everyday language, eliminating the need for specific commands or technical jargon. This intuitive interface enhances user experience and accessibility, enabling seamless interaction across diverse demographics and skill levels.

2. **Personalized Assistance:**

AI automation leverages data analytics and user insights to deliver personalized assistance tailored to individual preferences and needs. By analyzing user behavior, preferences, and past interactions, these assistants can anticipate user intent and provide relevant recommendations, suggestions, and solutions in real-time.

3. **Efficiency:**

AI automation helps streamline tasks and workflows by automating routine processes and simplifying complex tasks. From scheduling appointments and managing emails to conducting research and generating reports, these assistants empower users to focus on high-value activities while delegating repetitive tasks to AI-driven automation.

4. **Continuous Learning and Improvement.**

AI Assistants employ machine learning algorithms to continuously learn from user interactions and feedback, improving their accuracy and performance over time. Through iterative training and refinement, these assistants evolve to better understand user preferences, context, and nuances, enhancing the quality and relevance of their responses.

5. **Integration and Connectivity:**

AI Assistants seamlessly integrate with a variety of digital platforms, applications, and devices, enabling cross-channel connectivity and interoperability. Whether accessing information from the web, interacting with third-party services, or controlling smart home devices, these assistants serve as central hubs for accessing and managing digital resources.

CHAPTER 2
DESCRIPTION OF ORGANIZATION

ELEMENTALS AI

Introduction

ElementalsAI is a company that focuses on advanced research and provides cutting-edge AI-powered solutions. The team consists of highly capable AI engineers who put the most advanced research into production, creating a bridge between research and real-world problem-solving. They empower companies with real-time data-driven insights through a customized blend of emerging technologies such as Machine Learning, Natural Language Processing & Computer Vision to achieve maximum business value, and enhance organizational decision-making.

ElementalsAI offers customized ML solutions, hassle-free data preparation, and easy deployment & scalability of cutting-edge Machine Learning models. They also provide strategic insights to keep track of your business progress and project developments.

In collaboration with esteemed international platforms like Educative & Datacamp, ElementalsAI proudly launches specialized AI courses tailored for the next generation of tech enthusiasts. They believe in “Knowledge sharing enables equal opportunity for all” and offer internships in various different fields such as AI and DevOps. They provide mentorship to help students and enable them to move forward. They also have an alternative channel open always to help when in need or when struck in any kind of unfavorable situations.

Founder

Suraj Donthi is the founder of ElementalsAI. He is a Deep Learning practitioner with experience in applying deep learning and machine learning algorithms to solve complex problems in various domains. His expertise spans across automotive, retail, surveillance, biomedical image processing, and trading as well as analytics.

In addition to leading ElementalsAI, Suraj Donthi is also a Deep Learning Engineer & Consultant. He has authored curriculum and created content on DataCamp, a popular platform for learning data science. His most recent course on the platform is “Dealing with Missing Data in Python” which has been taken by over 21,000 learners.

His work reflects his commitment to bridging the gap between research and real-world problem-solving, allowing businesses to utilize the best of AI solutions to scale up and improve their operations.

Know more about them on: https://elementals.ai/

HR Executive

Tarun Jith

Tarun Jith is an HR Executive at ElementalsAI. He is a passionate expert in Human Resource Management. His main focus area is Human Resource Management and he provides services to different categories of learners like passed-out students for IT and Non-IT Jobs. He believes in the mantra “Right kind of Job for the Right Candidates” and works towards achieving this goal.

As an HR Executive, Tarun Jith plays a crucial role in the organization. He is often the first point of contact for prospective employees. His responsibilities include crafting compelling job descriptions, conducting interviews, and ensuring a seamless onboarding process. He also handles employee relations and conflict resolution, implements HR policies and procedures, and manages HR operations.

Tarun Jith is dedicated to cultivating leadership within the organization and nurturing a culture where every employee is empowered to become a leader in their own right. He is committed to providing a dynamic and immersive learning environment, offering flexible learning options, and fostering a supportive community. His work at ElementalsAI reflects his commitment to managing, nurturing, and optimizing the most valuable asset of the organization – its people.

ABOUT THE COMPANY

<u>About ElementalsAI</u>

ElementalsAI is a company that focuses on advanced research and provides cutting-edge AI-powered solutions. The team consists of highly capable AI engineers who put the most advanced research into production, creating a bridge between research and real-world problem-solving.

VISION STATEMENT

A world of scalable automated AI agents.

MISSION STATEMENT

Automate repetitive tasks using AI and innovate on new technologies.

Pro-active Assurance

Their ML algorithms are extremely scalable and can be fine-tuned for automating most of the tasks.

Automated Solutions

Their InstantRecruit platform aims to automate the entire Recruitment life cycle from shortlisting to technical interviews.

<u>Internship</u> - Positions available at ElementalsAI for the following Profiles when applied through Naukri, Internshala and LinkedIn.

- DSA instructors
- Content Development
- Digital Marketing
- AI Research
- Business Development Executive

CHAPTER 3
SCOPE OF WORK

PROBLEM STATEMENT

Developing an AI Automation Agent to generate contextually coherent articles:
In response to the growing demand for high-quality, contextually coherent content across various digital platforms, there emerges a compelling need to develop an AI Automation Agent. This AI Automation Agent will serve as an advanced solution to generate articles that are contextually coherent by going through blogs, research papers, and other relevant sources of information. By leveraging the capabilities of artificial intelligence and natural language processing, this initiative aims to automate the content generation process, enhance the quality of the produced articles, and ensure their contextual relevance. The development of such an AI Automation Agent represents a significant stride towards achieving content excellence and ensuring the seamless creation of high-quality articles in today's fast-paced digital environment. This will not only save time and resources but also provide a scalable solution to meet the ever-increasing demand for quality content.

Introduction to Basic Assistant System

AI automation agents, also known as AI bots or AI assistants, are software programs designed to automate a variety of tasks that would typically require human intelligence. These tasks can range from simple ones like setting reminders or answering queries to more complex ones like generating contextually coherent articles or managing network infrastructures.

At their core, AI automation agents leverage technologies such as Machine Learning (ML), Natural Language Processing (NLP), and Data Analytics to learn from data, understand human language, and derive insights respectively.

In today's digital age, AI automation agents are becoming increasingly prevalent and are being used in a variety of fields, including customer service, healthcare, marketing, and content creation, to name a few. They represent a significant step towards achieving operational efficiency and enhancing user experience.

However, it’s also important to consider the ethical implications and potential risks associated with their use, such as data privacy and job displacement.

Through leveraging the OpenAI platform, I gained invaluable insights into the potential of natural language processing and machine learning in optimizing user interactions and automating various operations. By harnessing the capabilities of FastAPI, I learned to develop robust and scalable APIs, enabling seamless communication between frontend and backend systems. Additionally, my exploration of React.js equipped me with the tools to design responsive and intuitive user interfaces, elevating the overall user experience of the AI Assistant. Throughout this journey, I not only mastered the intricacies of these technologies but also gained a profound understanding of web development processes, from conceptualization to deployment. By immersing myself in the development of this AI-powered assistant, I honed my skills in problem-solving, project management, and collaboration, laying a solid foundation for future endeavors in the dynamic landscape of AI-driven solutions and web development. This internship served as a catalyst for my professional growth, empowering me with the knowledge and expertise to navigate the intersection of artificial intelligence and web development with proficiency.

CHAPTER 4
LEARNING OBJECTIVES

1. **Scalable AI Solutions:**

ElementalsAI aims to empower businesses with scalable AI solutions. Interns will learn about the complete AI Model Development Lifecycle, from data preparation & labeling to deploying a scalable AI infrastructure.

2. **Customized ML Solutions:**

Interns will gain hands-on experience in leveraging structured Machine Learning pipelines to train complex data models and develop fully automated and insightful outputs.

They will learn how to elevate business understanding with text analytics and deploy advanced AI assistants for information extraction and automation of various tasks.

3. **Research-Oriented Learning:**

ElementalsAI provides an opportunity for interns to participate in cutting-edge research. This includes areas like Model Compression, Self-Supervised, Semi-supervised & Weakly Supervised Learning, Graph Neural Networks, Probabilistic Graphical Models, and Federated Learning.

Interns will learn how to employ futuristic technologies to draw strategic insights and optimize business operations. Interns will get a chance to kickstart their projects in no time with a blueprint and a product roadmap provided by ElementalsAI.

4. **Research on Large Language Models (LLMs):**

Interns will engage in research and exploration of Generative AI Open Source Language Models (LLMs), focusing on understanding their capabilities, limitations, and applications in natural language processing (NLP) and text generation tasks. Specifically, interns will investigate Gen AI LLMs, such as GPT-3.5 and its variants, exploring their architecture, training methodologies, and performance characteristics.

These learning objectives align with ElementalsAI's mission to empower companies with real-time data-driven insights and enhance organizational decision-making.

CHAPTER 5
CHALLENGES AND SOLUTION

In the development of an AI agent with a focus on generating coherent content without hallucinations, several non-functional challenges emerge alongside the functionalities. While meeting user requirements is essential, ensuring accuracy, maintainability, reliability, and security of the system are equally critical aspects of software engineering.

1. **Conformance to specific standards**: The AI agent must conform to industry standards and best practices in natural language processing and database management to ensure consistency and interoperability with existing systems and applications.

2. **Performance constraints:** The system should be optimized to handle large volumes of data and complex queries efficiently, without compromising on response times or system performance.

3. **Hardware limitations**: Consideration must be given to the hardware resources available for running the AI agent, ensuring compatibility and scalability across different hardware configurations.

4. **Maintainable**: The AI agent's codebase should be well-structured and modular, allowing for easy maintenance, updates, and enhancements over time.

5. **Reliable**: The system must be robust and resilient, capable of handling errors, failures, and unexpected inputs gracefully, while maintaining data integrity and system availability.

6. **Testable**: Adequate testing mechanisms should be in place to verify the accuracy and reliability of the AI agent's content generation functionality, including unit tests, integration tests, and end-to-end testing scenarios.

Solution:

To address accuracy issues in generating SQL queries related to networking terms, the Retrieval-Augmented Generation (RAG) model offers a promising solution. RAG combines retrieval-based and generation-based approaches, allowing the AI assistant to retrieve relevant context from a knowledge source (such as a database of networking terms) and generate SQL queries based on this retrieved context. By leveraging pre-trained language models and fine-tuning them on domain-specific data, RAG can accurately capture the nuances of networking terminology and generate contextually appropriate article generation.

Additionally, RAG's ability to learn from user feedback and adapt to different query contexts further enhances its accuracy and effectiveness in generating content related to a particular outline. Integrating RAG into the AI agent’s architecture can significantly improve the accuracy and reliability of its content generation functionality, thereby addressing one of the key challenges in its development.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

- Front End: NextJS
- Data Base: PostgreSQL
- Server: E2E server (EC2)
- API: OpenAI assistants

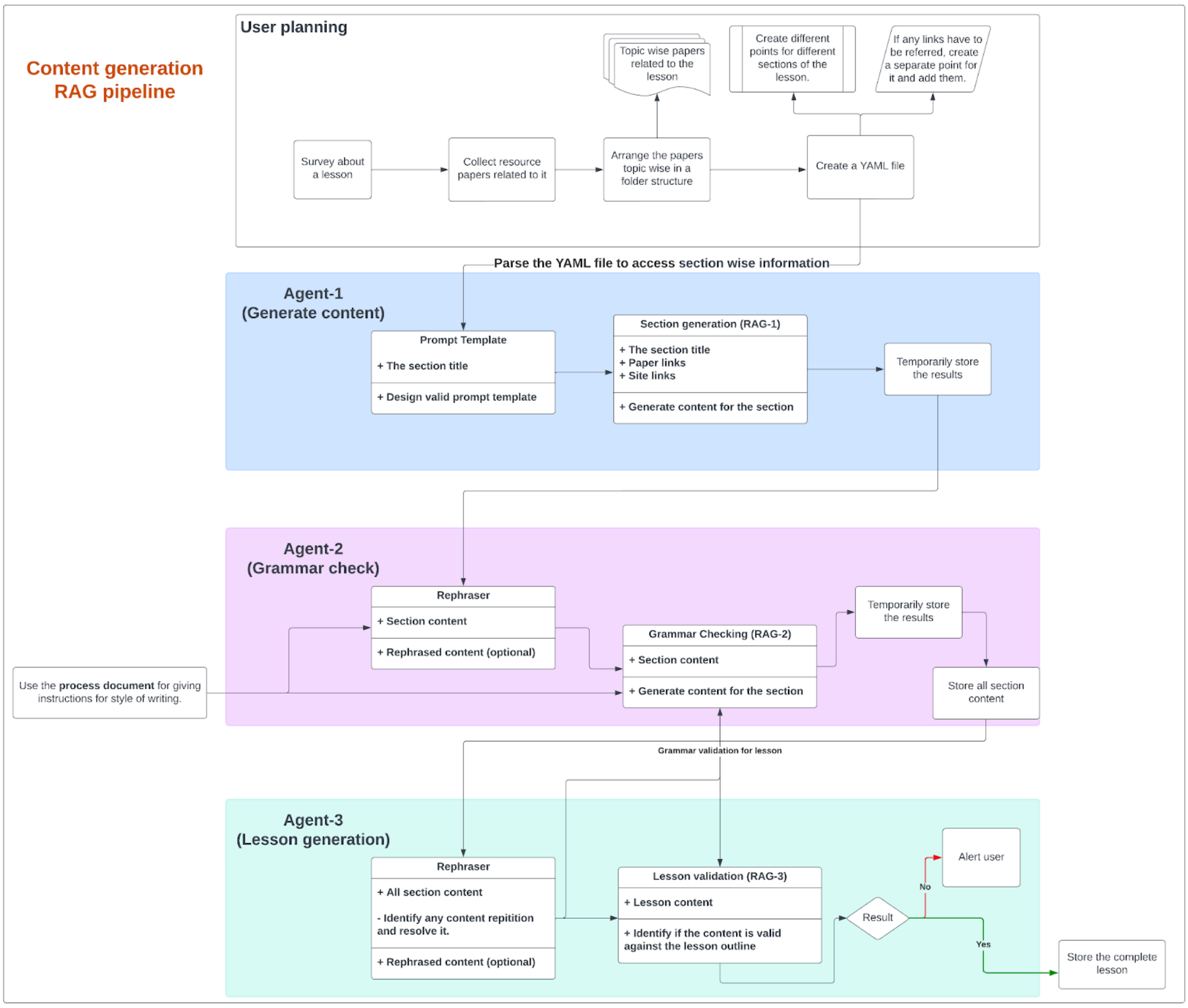
Hardware Requirements:

- VS Code
- Processor: Intel core i5 processor
- Memory: 8 GB
- Hard Disk: 20 GB

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

<u>System Design:</u>



The AI Agent System mainly contains:

- **RAG(Retrieval Augmented Generation)** - RAG is a state-of-the-art model developed by OpenAI that combines the capabilities of retrieval-based and generation-based approaches in natural language processing. Retrieval-based methods involve retrieving relevant information from a large corpus of text based on user queries, while generation-based methods focus on generating responses from scratch.
- **Embeddings** - Embeddings are dense vector representations of words or phrases in a

continuous vector space. These representations capture semantic relationships between words and enable machines to understand and process natural language more effectively. We used embeddings mainly in prompting to retrieve required documents for the content generation process.

- **AI Assistant** - AI Assistant is a multifaceted tool designed to streamline outline-to-article generation, offering users seamless access to valuable insights from complex datasets. Leveraging advanced natural language processing techniques, it interprets user queries, converting them into structured queries for document retrieval. Once executed, the assistant meticulously analyzes the results, extracting and combining various documents, identifying patterns, and curating high quality summarized content. By automating these processes, the assistant enhances productivity, reduces manual effort, and empowers users to make informed decisions based on data-driven intelligence

Flow of the process

Article Outline > Embeddings > Similarity Search (FAISS) > Few Shot Prompting > Feedback loop > Retrieved Data > Tree of Thought > Content Generation

During my internship, I spearheaded the development of an advanced AI agent tailored specifically for reporting tasks within the organization. This AI agent represented a significant milestone in enhancing reporting efficiency and accuracy, leveraging cutting-edge technologies in natural language processing and data analytics. By harnessing the power of machine learning algorithms and advanced language models, I engineered a solution capable of automating report generation, analyzing complex datasets, and extracting actionable insights from large volumes of data.

CHAPTER 7

REFLECTION AND ANALYSIS

My internship experience provided me with invaluable insights and learning opportunities, allowing me to gain a deeper understanding of key concepts and technologies in the dynamic landscape of networking and artificial intelligence (AI). Working in a startup environment presented unique challenges and opportunities, fostering a culture of innovation, collaboration, and rapid iteration. Through hands-on experience and mentorship, I honed my skills in network basics, explored various implementations of Large Language Models (LLMs), developed agents using LangChain and LlamaIndex, and gained proficiency in deploying applications on deployment strategies.

Additionally, my exploration of various implementations of LLMs opened my eyes to the transformative potential of AI in networking operations. From automating documentation and generating section-wise content to enhancing user interactions with AI-powered agents, I witnessed firsthand how AI-driven solutions can streamline processes, improve efficiency, and drive innovation in creative content generation.

Finally, deploying applications on E2E servers provided me with valuable insights into the deployment and management of web applications in production environments. By configuring the instances for load balancing, caching, and SSL termination, I learned how to optimize performance, laying the groundwork for scalable and resilient deployments.

In conclusion, my internship experience was a transformative journey that allowed me to deepen my knowledge, expand my skill set, and gain practical experience in networking, AI, web development, and DevOps practices. Through reflection and analysis, I have identified areas of growth and development, paving the way for continued learning and professional advancement in the dynamic field of technology.

CHAPTER 8
RECOMMENDATIONS

Flexible Training Schedule:

Offer interns the flexibility to choose their working hours, accommodating their individual preferences and schedules. By allowing interns to work during times that suit them best, you empower them to maintain a healthy work-life balance and optimize their productivity.

Research Oriented approaches:

Encourage interns to participate in research-oriented projects that align with their academic interests and career goals. These projects not only provide a practical application of their theoretical knowledge but also expose them to the latest trends and advancements in the field. By working on such projects, interns can contribute to the development of innovative products and solutions, thereby gaining valuable experience and enhancing their problem-solving and critical thinking skills. This hands-on experience with research-oriented projects can be instrumental in fostering a culture of innovation and continuous learning.

Mentorship Program:

Implement a mentorship program that pairs each intern with an experienced mentor who can provide guidance, feedback, and support throughout their internship. Mentors play a crucial role in facilitating learning, fostering professional growth, and helping interns navigate challenges effectively. Encourage mentors to schedule regular check-ins with their mentees and provide ongoing support and encouragement.

Feedback and Evaluation:

Provide interns with regular feedback and evaluation sessions to assess their progress, identify areas for improvement, and recognize their achievements. Feedback sessions should be constructive, specific, and actionable, highlighting interns' strengths and areas for development. Encourage interns to reflect on their experiences, set goals for their learning journey, and take ownership of their professional development. By fostering a culture of continuous feedback and improvement, you empower interns to maximize their potential and make meaningful contributions to the company.

CHAPTER 9
CONCLUSION

- This application provides a powerful tool for providing insights to users, automating the reporting process, and conducting performance analysis.
- This application also helps in retrieving relevant content from a research paper, blogs, articles from the internet.
- This software helps in reducing the clerical work of visiting the papers repeatedly for various queries and taking the aggregation part out of context.
- This system provides fast, efficient, reliable and user-friendly interfaces in the chat system and has no chance of losing data or any write operations while processing the user query during the content generation.
- This software provides a good user interface such that a user of basic computer knowledge can operate the application. It also reduces effort done by converting a YAML based lesson outline to a fully fledged, user-friendly lesson.

CHAPTER 10
REFERENCES

1. Founder : Suraj Donthi

   https://www.linkedin.com/company/elementalsai/

2. HR Executive : Tarun Jith K
   https://in.linkedin.com/in/tarun-jith-k-90874433